Nader Sghir

Lead Data Engineer | Data & AI Platforms

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Summary

Lead Data Engineer with 16 years building production data platforms that feed ML and optimisation models, from airline demand forecasting to aircraft predictive maintenance and multilingual enterprise search. Hands-on across Databricks/Spark on Azure, Airflow and Athena on AWS, and BigQuery. Python-first engineer applying best practices (typing, linting, pre-commit, CI) and AI-assisted development with Anthropic coding agents. Owns the full path from raw data to versioned datasets, MLflow-tracked models, REST APIs and user-facing apps. Contributes to IVADO Labs' internal Python kickstart library, including LLM fine-tuning. Bilingual EN/FR.

Core Skills

- **Languages:** Python (pandas, Polars, PySpark, Pydantic, FastAPI, SQLAlchemy), SQL (advanced, query optimisation), Scala, Java
- **Data platforms:** Databricks, Apache Spark, Snowflake, Airflow, Azure Data Factory, Talend
- **Cloud:** AWS (S3, Glue, Athena, Secrets Manager), Azure (Databricks, ADF, Cosmos DB), GCP (BigQuery)
- **ML enablement:** MLflow (training runs, model versioning), training and feature datasets, inference-serving APIs, embedding evaluation, LLM fine-tuning (PEFT, TRL)
- **Engineering:** Typed, linted Python (mypy, Ruff, Pydantic), pre-commit hooks, GitLab CI/CD, Terraform (IaC), Docker, Poetry/uv, REST API design (Apigee)
- **Data quality and governance:** schema drift detection, freshness validation, sparsity/flatline checks, dataset versioning and registry
- **AI-assisted development:** Anthropic Claude coding agents for implementation, refactoring and tests, under typed, linted, CI-gated workflows
- **Applications:** role-based access and authentication, human-in-the-loop decision UIs
- **Databases:** PostgreSQL, SQL Server, Oracle, MongoDB, Cosmos DB, Elasticsearch

Experience

IVADO Labs Jul 2019 – Present
Lead / Senior Data Engineer · Client delivery for AI and data platforms *Montréal, QC*

Enterprise Search — AI-Powered Image & Multilingual Search *(AWS, Snowflake)* 2024 – Present

- Built and maintain REST APIs serving image and multilingual search results in production (1,000+ requests/day, 2 languages), integrating encoder outputs into search infrastructure.
- Built a reusable encoder evaluation job for image and text embeddings. It selected the best encoder for the use case and is rerun to benchmark encoders on each new customer catalog.
- Store and process customer catalog data in Snowflake; provision infrastructure with Terraform.
- Mentor and onboard engineers; set documentation standards and code-review practice for a 3-person data team.

Travel & Hospitality — Data Platform for ML Revenue Optimisation *(AWS)* 2023 – 2024

- Built Airflow-orchestrated pipelines moving raw data from GCP BigQuery into S3 Parquet, queried through Glue/Athena, with credentials managed in AWS Secrets Manager.
- Migrated orchestration from Python/Lambda scripts to Airflow. Designed a Medallion layout (Raw → Bronze → Silver → Gold) using parameterised SQL, with full, incremental and partition-overwrite run modes.
- Introduced semantic versioning of Gold datasets with an S3 run registry and active-version pointer, so ML teams consume stable datasets while upstream pipelines evolve (3 models, 2 partners).
- Built data quality checks (freshness, schema drift, alerting): 99%+ pipeline uptime and SLA compliance. Provisioned infrastructure with Terraform.

Airline — Demand Forecasting Platform *(Azure)* 2021 – 2023

- Designed and operated ELT pipelines on Databricks (PySpark) with Azure Data Factory orchestration over 6TB+ of transactional data, feeding forecasting and operations-research models. Lab project on 20 origin-destination markets, validated, then scaled to more markets.
- Served ML/OR outputs from Cosmos DB to the planning UI.
- Established CI/CD and Git workflows for reproducible, auditable releases.

Aerospace — Aircraft Predictive Maintenance 2020 – 2021

- Built ELT pipelines for business-jet sensor telemetry, with automated quality gates (sparsity, flatline, out-of-range) for certified model training data.
- Managed model training runs and model versioning with MLflow; provisioned infrastructure with Terraform.
- Developed inference-serving REST API (FastAPI, SQLAlchemy) delivering model outputs to the operational UI.
- Built metadata and dependency-tracking framework for reproducibility and auditability of anomaly-detection datasets.

Port Logistics — Operational Analytics 2019 – 2020

- Designed data models and pipelines for vessel ETA prediction and operational forecasting.
- Built the full application end to end: UI, user management with roles and authentication, and human-in-the-loop decision making over model outputs.

Internal — Kickstart Python Library Ongoing

- Contribute to IVADO Labs' shared project starter (Polars, MLflow, FastAPI helpers, ML training loop); extending it with an LLM fine-tuning module (PEFT, TRL, training recipes).

Facilite Inc. / CGI Aug 2016 – Jun 2019
Senior Software & Data Engineer *Montréal, QC*

- Led data integration and ETL architecture for retail, manufacturing and fintech clients using AWS, Talend, Apigee and SQL.
- Migrated a fintech client's pipelines from Talend ETL to Airflow.
- Integrated ActiveMQ with Talend ESB to sync inventory across a retailer's stores and warehouse.
- Built real-time and batch pipelines integrating REST APIs and third-party sources under strict quality and compliance standards.

2i Intelligence Apr 2014 – Jun 2016
ETL & Database Designer *France*

- Designed dimensional warehouses and data marts for market-research firms.

TELNET Holding / Decade Dec 2010 – Mar 2014
Software Developer *Tunisia*

- Delivered a one-shot ERP data migration for an environmental-services group with zero customer impact.

Education

Engineering Degree, Computer Science (Software Engineering) — ENIS, Sfax, Tunisia 2010

Certifications

Databricks Generative AI Fundamentals (2024)

Languages

French (native) · English (full professional) · Arabic (native)